

Short positions allowed

Expected growth rate, $\mathbb{E}[g_p]$ (yr^{-1})

$\mathbb{E}[g_{p_1}]$

$\mathbb{E}[g_{p_2}]$

$\mathbb{E}[g_{p_3}]$

Efficient frontier

1

2

Global minimum
variance

Same risk,
higher expected growth

3

Lower branch

σ_{g,p_2}

$\sigma_{g,p_1} = \sigma_{g,p_3}$

Growth-rate standard deviation, $\sigma_{g,p}$ (yr^{-1})

